

Sho Sakaue

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EDUCATION

University of Utah

Bachelor of Science in Computer Science and Applied Mathematics

Salt Lake City, Utah

Aug. 2024 - Expected Dec. 2027

- **GPA:** 3.97/4.0
- **Relevant Coursework:**
 - * **CS:** Machine Learning, Deep Learning, Probabilistic Deep Learning, Natural Language Processing, Computer Vision, Computational Geometry, Data Structures & Algorithms, Models of Computation
 - * **Math:** Optimization, Probability Theory & Statistics, Linear Algebra, Numerical Analysis, Complex Analysis, Real Analysis, Calculus I-III
 - * **Large Language Models:** - Matsuo Lab, University of Tokyo

June. 2026 - Sep. 2026

WORK EXPERIENCE

Computer Science Teaching Assistant - CS 1400 (Intro to Computer Programming) July. 2026 - Present

University of Utah

Salt Lake City, Utah

- Led weekly group lab session to help students practice coding fundamentals and clarify concepts beyond lectures.
- Graded coding assignments for 100+ students based on correctness, readability, and programming best practice.
- Assisted students in debugging programs and developing their understanding through office hours.

Mathematics Tutor and Grader

University of Utah

Aug. 2025 - July. 2026

Salt Lake City, Utah

- Tutored students in Calculus I, II, III, Linear Algebra, and Statistics through clear explanations for each student.
- Worked with 10+ professors to grade 1,000+ student submissions, providing detailed feedback.
- Adapted strategies to each student's learning style to improve understanding and problem-solving independence.

PROJECTS

BayesBench | *numpy, scikit-learn, matplotlib, pandas*

June. 2026 - Present

- Implemented the benchmark of Bayesian Optimization method for various limited-budget black-box optimization.
- Added experiment scripts and plotting tools to compare optimization performance across methods.
- Structured the project as a reusable Python package for future extension to more benchmarks and optimizers.

LoRA and QLoRA for Fine-Tuning (Paper reproduction) | *PyTorch, Hugging Face*

Mar. 2026 – Apr. 2026

- Reproduced the LoRA and QLoRA methods from their original research papers, translating the proposed low-rank adaptation and quantized fine-tuning techniques into a working training pipeline.
- Fine-tuned a llama-2-1B and benchmarked LoRA and QLoRA against the base model and full fine-tuning, comparing accuracy-performance tradeoffs.
- Conducted hyperparameter experiments across LoRA ranks and quantization bit widths to analyze their effects.

Multiplayer Networked Game & HTTP Server - *Software Practice* | *C#, SQL*

Nov. 2025 - Dec. 2025

- Developed a distributed multiplayer game client in Blazor using the MVC architectural pattern.
- Implemented a multi-threaded TCP network controller to communicate by JSON, using locks to ensure thread safety and prevent race conditions.
- Integrated a MySQL database to store game history and player high scores.

TECHNICAL SKILLS

Programming Languages : Python, C/C++, Java, Bash, SQL(MySQL), C#

Markup & Style Languages: HTML, CSS, $\text{\LaTeX}$, Markdown

Machine Learning: PyTorch, NumPy, Hugging Face, scikit-learn, SciPy, Matplotlib, Pandas

Developer Tools: Git(GitHub, GitLab), AWS Cloud, Docker, Linux, HPC/SLURM computing

RESEARCH EXPERIENCE

- Undergraduate Research Assistant** - *under Professor Suhas Prameela* Aug. 2025 - Aug. 2026
University of Utah *Salt Lake City, Utah*
- Generated and pre-processed training datasets for Machine Learning (ML) models for atomic simulation.
 - Increased training convergence rate by day and model performance selecting optimal optimization method.
 - increased model prediction accuracy by 15 % on average while maintaining inference speed.

CONFERENCE & PRESENTATION

- Undergraduate Research Symposium** - University of Utah July. 2026
University of Utah
- Delivered a presentation on research funded through the Undergraduate Research Opportunity Program (UROP).
 - Presented research on developing *machine learning interatomic potentials for NiCoCr-based alloys*.

LEADERSHIP EXPERIENCE

- IBM Quantum Qiskit Fall Fest 2026 Co-Organizer** sponsored by IBM Quantum July. 2026 - present
University of Utah *Salt Lake City, Utah*
- Invited Professors who is working in different aspect of Quantum Computing to give broad insights to attendees
 - Working as a marketer and made Instagram account to promote this event for campus wide
- UJC(Utah Japanese Community) officer** Aug. 2025 - present
University of Utah *Salt Lake City, Utah*
- Collaborated with TKF Project Inc. and organized career support event for more than 40 people
 - Organized community event, and gathered students from various background

COMPETITION/HACKATHON

- MIT iQuHACK** (Quantum Computer) Feb. 2026
- Simulated 50–60 qubit quantum circuits using Qiskit to identify hidden peak bitstrings, placing 24th in BlueQubit Challenge.
 - Optimized quantum circuits simulation performance by gate reordering, reducing runtime by approximately 30%.
 - Developed statistical candidate-selection heuristics to infer peak from near-uniform output distributions.
- Crimson Hack** (University of Utah) Oct. 2025
- Built a rocket simulation game, where players navigate from one planet to another while avoiding obstacles.
 - Implemented planet-specific gravity and fuel-limited movement mechanics considering physical reality.
 - Designed interactive gameplay logic that allowed users to select different solar-system planets and control rocket

AWARDS & CERTIFICATES

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| Frank Stenger Scholarship | Fall 2026 |
| Undergraduate Research Opportunity Program (UROP) | Summer 2026 |
| 24 th place in BlueQubit challenge in MIT iQuHack | Feb. 2026 |
| Dean's List | Fall 2024 - Summer 2026 |
| ISSS Scholarship | Spring 2026 |
| Utah Global Scholarship | Fall 2024 - Fall 2027 |

LANGUAGE

- English** - Fluent
Japanese - Native